IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-IT-031

1. Topic (12 words max)

Smart Water Logging ,Flood Prevention and Post Flood Event Management System

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Our project aims to address the recurring loss of lives and property caused by flood events, which frequently affect communities in vulnerable regions of the country. These floods often occur unpredictably and can extend beyond expected geographical boundaries, leaving residents with little or no time to evacuate, secure their valuables, or assist vulnerable individuals. As a result, significant human and economic losses are incurred, severely disrupting daily life and livelihoods within these communities each time such an event occurs without adequate warning.

6. Brief description of Existing Research and Systems (300 words max)

Several commercial products and pilot implementations currently address drainage monitoring and flood risk management using IoT technologies. Solutions such as Smart Drain systems deploy networked ultrasonic and cloud-connected sensors to monitor stormwater pits and drainage channels in real time. These systems provide early alerts to maintenance teams, reduce the need for manual inspections, and support decision-making through dashboards and data APIs. Similarly, collaborative initiatives such as the Smart Stormwater Drains project by Downer, Fujitsu, and EYEfi utilize sonar and camera-based sensors to monitor water levels and flow conditions in urban drainage infrastructure. These systems focus on early warning alerts, remote monitoring, and improved response times to reduce flood risks in cities. Alongside commercial systems, academic research has extensively explored IoT-based drainage and flood management solutions. Studies such as IoT-Based Detection of Blockages in Stormwater Drains propose the use of ultrasonic and environmental sensor nodes communicating via LoRaWAN to cloud platforms to limitations remain. Most existing systems primarily focus on monitoring and alert generation rather than fully automated mitigation mechanisms such as intelligent flow control or dynamic water redirection. Many solutions are reactive in nature and lack predictive analytics capable of forecasting future blockages or flood risks using historical and environmental data. Furthermore, several systems do not provide actionable decision-support dashboards tailored for municipal authorities, instead presenting raw sensor data without prioritization or recommended actions. Practical challenges related to communication range, sensor cost, long-term scalability, and power autonomy, including limited use of solar energy, further restrict widespread real-world deployment.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

• The proposed Smart IoT Drain Blockage Management and Flood Reduction System with Flood Prediction Map directly addresses the gaps identified in Section 6 by providing an integrated, predictive, and automated solution. Existing systems focus primarily on monitoring and alerting, lack predictive analytics, and do not integrate actionable dashboards or automated mitigation mechanisms. This system resolves these limitations by combining real-time drain blockage detection, water-level monitoring, predictive flood risk mapping, and automated mitigation. Historical flood and rainfall data, along with real-time sensor inputs, are used to forecast potential blockages and identify high-risk zones, enabling proactive decision-making. The inclusion of a Flood Risk Map categorizing areas into High Threat, Medium Threat, and Low Threat zones provides municipal authorities with actionable insights and helps prioritize interventions. Additionally, the use of Arduino-based low-cost sensors with optional solar-powered modules ensures scalability, power autonomy, and suitability for local urban conditions.

Beyond flood prevention, the system is extended into post-flood response and recovery. A post-flood disease management component is introduced, which leverages symptom-based inputs combined with facial scan analysis to assist in the early identification of common flood-related diseases. This enables quicker medical intervention and reduces the risk of disease outbreaks in affected communities. Furthermore, a post-flood ration and rescue management feature is integrated, allowing affected individuals or field officers to input victim location data through the system. This information is visualized on the dashboard, enabling relief workers and volunteers to accurately locate victims, prioritize rescue operations, and distribute food, water, and essential supplies efficiently.

• From a system perspective, the solution follows a clear Input–Process–Output (IPO) model:

Inputs:

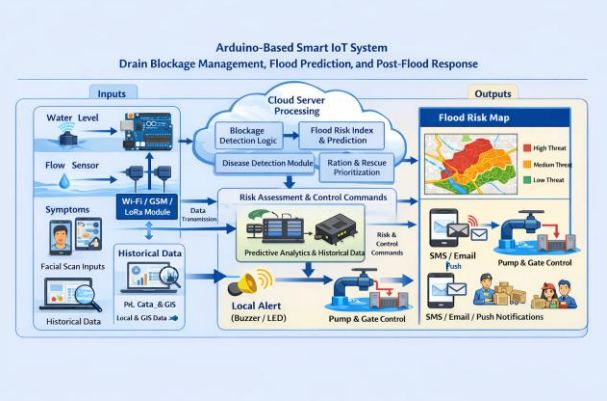
• Real-time data from ultrasonic water-level sensors (upstream and downstream), flow sensors, and optional environmental sensors.
• Rainfall data from IoT rain sensors or local weather APIs.
• Historical flood data and topography information for each area.
• Post-flood symptom data and facial scan inputs for disease detection.
• Victim location and status data entered via mobile or web interfaces.

Process:

• Sensor data is collected by an Arduino microcontroller and transmitted to a cloud server via Wi-Fi, GSM, or LoRaWAN.
• Blockage detection logic calculates water-level differences and triggers alerts when thresholds are exceeded.
• Flood Risk Index is computed for different zones using a weighted combination of sensor readings, rainfall intensity, and historical data.
• Predictive analytics forecast areas likely to be affected by flooding.
• Post-flood health analysis processes symptom data and facial scans to suggest potential disease risks.
• Location data is processed to generate rescue and ration distribution priorities.

Outputs:

• Local alerts via buzzer or LED for detected blockages.
• Real-time dashboard displaying water levels, blockage status, flood threat zones, victim locations, and health risk indicators.
• Automated activation of servo gates or pumps for water redirection.
• Flood Risk Map with color-coded zones: Red for High Threat, Orange for Medium Threat, and Green for Low Threat.
• Disease risk alerts and health recommendations for post-flood scenarios.
• Notifications to municipal authorities, medical teams, and relief workers via SMS, email, or push alerts.



8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The proposed Smart IoT Drain Blockage Management and Flood Prediction System can be successfully implemented using a combination of readily available domain expertise, technical knowledge, and accessible data sources. Specialized knowledge related to urban drainage systems, flood behavior, and water flow dynamics can be gained through an in-depth review of existing research publications, government flood management reports, and disaster mitigation guidelines. Studying prior case studies on urban flooding and drainage failures helps the project team understand real-world constraints and operational challenges. In addition, guidance from academic staff in civil engineering, environmental science, or disaster management can further strengthen the team's understanding of drainage infrastructure and flood risk assessment.

The technical knowledge and skills required for the project, such as Arduino programming, sensor integration, wireless communication, and automation, are widely available through open-source platforms and online learning resources. Team members can gain hands-on experience by working with ultrasonic water-level sensors, flow sensors, servo motors, and communication modules such as Wi-Fi, GSM, or LoRa. Practical experimentation and prototyping will allow members to validate system logic, calibrate sensors, and test automated responses. Skills related to cloud integration, data storage, and dashboard development can be developed using widely used technologies such as Firebase or web-based visualization tools. Basic data analysis and predictive modeling techniques can be learned through tutorials and applied directly to collected sensor and rainfall data.

The data required for this project is obtained from both real-time and historical sources. Real-time data is collected directly from sensors installed in upstream and downstream drain locations, measuring water levels, flow variations, and potential blockage conditions. Rainfall intensity data can be gathered using rain sensors or sourced from publicly available meteorological datasets. Historical flood records, rainfall patterns, and geographic information needed for flood risk mapping can be accessed from open government data portals, disaster management agencies, and GIS platforms. These datasets enable the classification of affected areas into high, medium, and low flood risk zones, supporting predictive analysis and visualization.

This project does not involve human subjects, personal data, or sensitive information, as it focuses solely on environmental monitoring and infrastructure analysis. Therefore, formal ethical clearance is not strictly required. However, ethical best practices such as responsible data handling, transparency in system outputs, and ensuring that the system does not interfere with public safety or infrastructure will be maintained. Any future real-world deployment involving local authorities or public infrastructure will be carried out with appropriate permissions and compliance with regulatory requirements.

9. Objectives and Novelty

Main Objective

The main objective of this project is to develop a low-cost Arduino-based Smart IoT system that detects drain blockages, predicts flood-prone areas, and reduces flood impact through automated mitigation and decision support. The system integrates real-time and historical data to generate flood risk maps and also supports post-flood disease detection and rescue and ration management to improve disaster response and recovery.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Somarathne B.D. IT22091734	Flood map creation	Take data from the IOT sensers such as the height of the location using domestic iot sensors installed in the base level of houses and using the predicted water levels using forecasted data create a geo flood map which marks the areas which will be affected based on the threat level. Areas will be marked as high threat, medium and low threats marked by colors on the map.	The proposed system utilizes real-time and historical data to generate predictive insights ahead of potential flood events, rather than relying solely on immediate real-time conditions. The map allows residents to make evacuation plans to nearby threat free areas. By forecasting risk scenarios in advance, the system extends the available response time, enabling authorities and communities to prepare earlier, implement preventive measures, and reduce the overall impact of flooding.
IT22278258 Bulumulla D.M.D.B	Drain water level prediction Real-time drain monitoring and alert system	Collect real-time data from IoT sensors in drainage systems, including water level, flow rate, rainfall intensity, and blockage indicators, and integrate it with historical datasets to support predictive analysis. The system processes this data to identify abnormal conditions, assess flood and blockage risks, and forecast affected areas and timelines. Results are visualized through a web-based dashboard with interactive maps, while automated alerts and notifications are generated when predefined thresholds are exceeded.	This approach integrates IoT-based real-time monitoring with data-driven predictive modeling to forecast drain water levels in advance. Unlike traditional reactive systems, the solution provides early warnings with location- and time-specific impact estimation, enabling proactive drainage management and timely preventive actions to minimize flood risks.

Jayanath H.P.K.N IT22170002	To design and implement a post-flood disease management module that enables early detection of flood-related health risks using symptom data and facial scan analysis.	Collect symptom-based inputs from affected individuals through a web or mobile interface; capture facial scan data to identify visible indicators such as fatigue, skin discoloration, or swelling; process inputs using rule-based or machine-learning models to estimate disease risk; generate health alerts and recommendations; visualize health risk indicators on the dashboard for medical teams.	Integrates flood management with post-disaster healthcare support by combining symptom reporting and facial scan analysis, enabling early disease risk detection and faster medical intervention—an aspect largely absent in existing flood monitoring systems.
Rashmi P.M.U IT22130198	To develop a post-flood rescue and ration management system that identifies affected individuals' locations and supports efficient rescue and relief distribution.	Enable users or field officers to enter victim location and status data; process and map affected locations on a real-time dashboard; prioritize rescue and ration distribution based on risk level and accessibility; notify rescue teams and volunteers; update victim status after rescue or aid delivery.	Extends traditional flood monitoring systems into disaster response and recovery by integrating real-time victim location mapping with rescue prioritization and ration distribution, improving coordination and efficiency during post-flood operations.

10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

Good to proceed with.

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Deemantha	Siriwardana	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				